Home Reading 05 — Frequency Response

Before Class

Preview Lecture 5

Read Lecture 5: Frequency Response.

Textbook Reading

- Required: Ogata, System Dynamics, 4th ed., Section 9-2, pp. 432–435 (sinusoidal transfer function, magnitude and phase).
- Page numbers above are printed textbook pages. Use the Lecture slides for the remaining worked examples.
- Reference only: Section 9-3, pp. 438–441, for rotating unbalance and vibration.

This Week Introduces

Sinusoidal response, Bode construction, resonance, bandwidth and the use of frequency-response data to validate a model.

Tutorial Preparation

Read Tutorial 5 before class. Be ready to:

- Q1: Interpret sinusoidal gain, phase and elementary Bode factors.
- Q2: Analyse a fan whose unbalance force changes with speed.
- Q3: Infer a numerator zero from Bode measurements.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- Why does the steady output of a linear system have the same frequency as a sinusoidal input?
- Why can rotating unbalance become more severe as rotational speed increases?

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